

Question 1:

$$H = \frac{Coul}{Cin} = \frac{100}{10} = 10$$

$$B = 8$$

1. NAND4 + INV

$$g = \frac{n+2}{3} = \frac{4+2}{3} = 2, \quad g_{INV} = 1.$$

$$P_{NAND4} = 4, \quad P_{INV} = 1$$

$$\therefore G = 2 \times 1 = 2$$

$$F = GBH = 2 \times 8 \times 10 = 160$$

$$f = (160)^{1/3} = 5.43$$

$$\therefore D = N \times f + P \\ = 3 \times 5.43 + 1 + 4 + 1 = \underline{22.29}$$

2. NAND2 + NOR2

$$G = \frac{4}{3} \times \frac{5}{3} = \frac{20}{9}$$

$$\therefore F = \frac{20}{9} \times 8 \times 10 = 177.78$$

$$f = (177.78)^{1/3} = 5.62$$

$$\therefore D = 3 \times 5.62 + 1 + 2 + 2 = \underline{21.86}$$

3. INV + NAND4 + INV

$$G = 1 \times 2 \times 1 = 2$$

$$GBH = 2 \times 8 \times 10 = 160$$

$$f = (160)^{1/4} = 3.56$$

$$D = 4 \times 3.56 + 1 + 1 + 4 + 1 = \underline{21.24}$$

4. NAND4 + INV + INV + INV

$$G = 2 \times 1 \times 1 \times 1 = 2$$

$$F = 2 \times 8 \times 10 = 160$$

$$f = (160)^{1/5} = 2.76$$

$$D = 5 \times 2.76 + 1 + 4 + 1 + 1 + 1 = \underline{21.8}$$

5. NAND2 + INV + NAND2 + INV + INV + INV

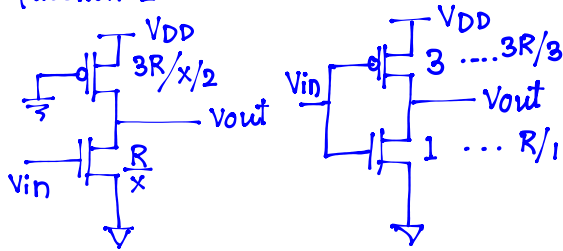
$$G = \frac{4}{3} \times 1 \times \frac{4}{3} \times 1 \times 1 \times 1 = \frac{16}{9}$$

$$F = \frac{16}{9} \times 8 \times 10 = 142.22$$

$$f = (142.22)^{1/7} = 2.03$$

$$D = 7 \times 2.03 + 1 + 2 + 1 + 2 + 1 + 1 + 1 = \underline{23.21}$$

Question 2:



$$\frac{V_{DD}/2}{R/x} - \frac{V_{DD}/2}{3R/x/2} = \frac{V_{DD}/2}{R}$$

$$\frac{x}{R} - \frac{x/2}{3R} = \frac{1}{R}$$

$$3x - \frac{x}{2} = 3$$

$$\frac{5x}{2} = 3$$

$$\therefore x = \frac{6}{5}$$

For inverter: NMOS = 6/5

$$\text{PMOS} = 6/10 = \frac{3}{5}$$

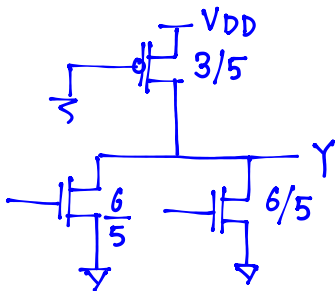
$$g_u = \frac{6/5}{\frac{3}{5} + \frac{1}{5}} = \frac{6}{5} \times \frac{5}{4} = \frac{6}{4} = \frac{3}{2}$$

$$g_d = \frac{6/5}{4} = \frac{6}{20} = \frac{3}{10}$$

$$P_u = \frac{\frac{6}{5} + \frac{3}{5}}{\frac{3}{5} + \frac{1}{5}} = \frac{9}{5} \times \frac{5}{4} = \frac{9}{4}$$

$$P_d = \frac{\frac{6}{5} + \frac{3}{5}}{4} = \frac{9}{5} \times \frac{1}{4} = \frac{9}{20}$$

NOR 2



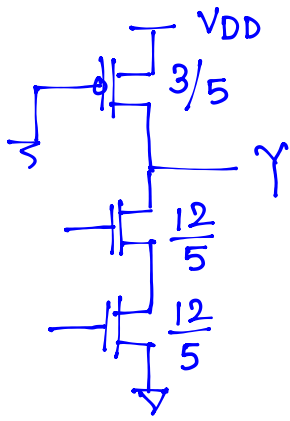
$$g_u = \frac{6/5}{\frac{3}{5} + \frac{1}{5}} = \frac{6}{5} \times \frac{5}{4} = \frac{6}{4} = \frac{3}{2}$$

$$g_d = \frac{6/5}{4} = \frac{6}{20} = \frac{3}{10}$$

$$P_u = \frac{\frac{6}{5} + \frac{6}{5} + \frac{3}{5}}{\frac{3}{5} + \frac{1}{5}} = \frac{15}{5} \times \frac{5}{4} = \frac{15}{4}$$

$$P_d = \frac{\frac{6}{5} + \frac{6}{5} + \frac{3}{5}}{4} = \frac{15}{5} \times \frac{1}{4} = \frac{3}{4}$$

NAND 2



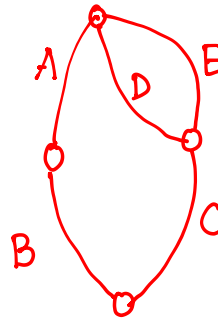
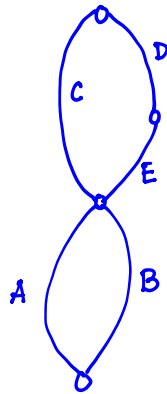
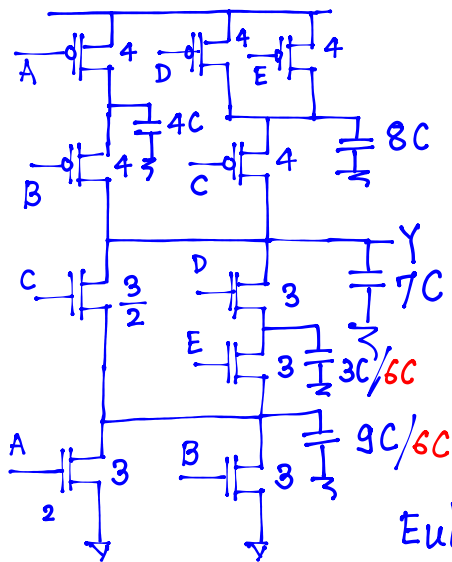
$$g_u = \frac{12/5}{\frac{3}{5} + \frac{1}{5}} = \frac{12}{5} \times \frac{5}{4} = 3$$

$$g_d = \frac{12/5}{4} = \frac{12}{5} \times \frac{1}{4} = \frac{3}{5}$$

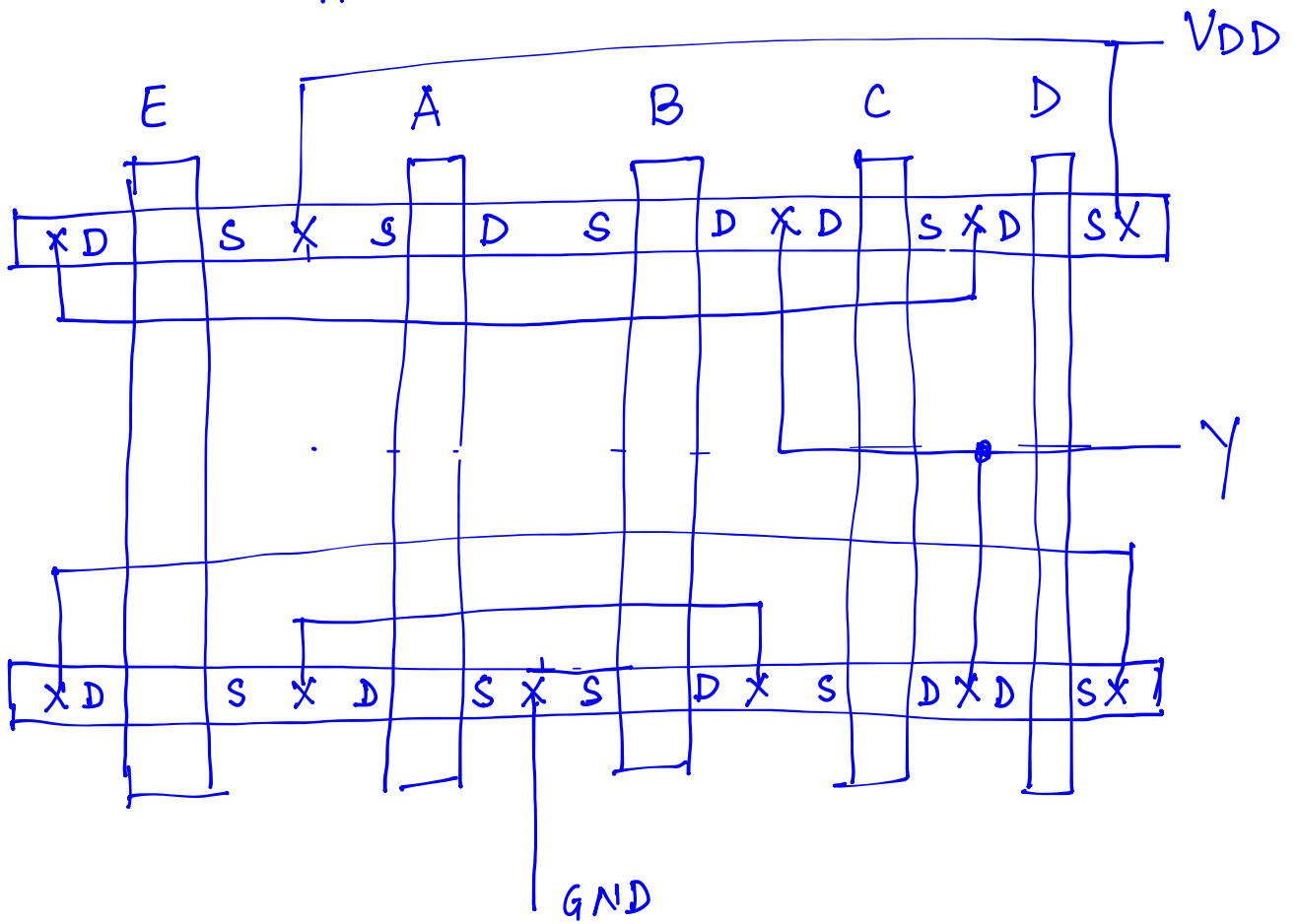
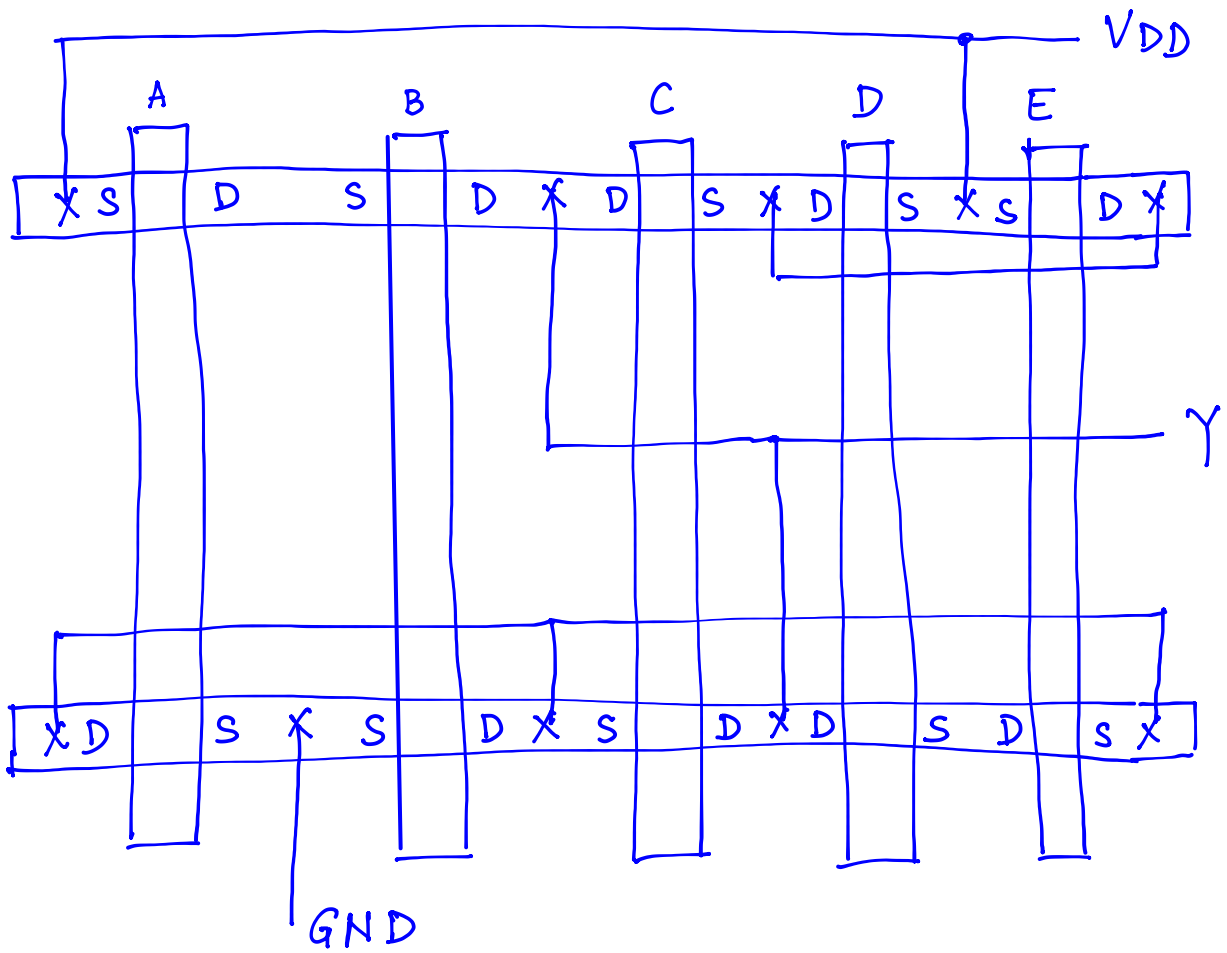
$$P_u = \frac{\frac{3}{5} + \frac{12}{5}}{\frac{3}{5} + \frac{1}{5}} = \frac{15}{5} \times \frac{5}{4} = \frac{15}{4}$$

$$P_d = \frac{\frac{3}{5} + \frac{12}{5}}{4} = \frac{15}{5} \times \frac{1}{4} = \frac{3}{4}$$

Question 3 :



Euler's path : ABCDE / EABCD



Question 4 : Setup calculations

FF1: At clk input : clock path delay(min) = 1 ns

At D input : max data path delay = 8 ns + setup of FF1 = 8 ns + 4 ns = 12 ns

$$\therefore \text{Setup time at B} = 12 \text{ ns} - 1 \text{ ns} = 11 \text{ ns}$$

FF2 Min clk path delay = 1 ns + 1 ns = 2 ns

For B input : max data path delay

$$= 8 \text{ ns} + 4 \text{ ns} + 7 \text{ ns} + 6 \text{ ns}$$

$$= 25 \text{ ns} + \text{setup time} = 25 \text{ ns} + 4 \text{ ns} = 29 \text{ ns}$$

$$\therefore \text{Setup at B} = 29 \text{ ns} - 2 \text{ ns} = \underline{\underline{27 \text{ ns}}}$$

For A input

$$\begin{aligned} \text{max data path delay} &= 6 \text{ ns} + \text{setup time} \\ &= 6 \text{ ns} + 4 \text{ ns} \end{aligned}$$

$$= 10 \text{ ns}$$

$$\therefore \text{setup at A} = 10 \text{ ns} - 2 \text{ ns} = \underline{\underline{8 \text{ ns}}}$$

For E input

$$\text{max data path delay} = 4 \text{ ns} + 7 \text{ ns} + 6 \text{ ns}$$

$$= 17 \text{ ns} + \text{setup} = 17 \text{ ns} + 4 \text{ ns} = 21 \text{ ns}$$

$$\text{set up at E} = 21 \text{ ns} - 2 \text{ ns} = \underline{\underline{19 \text{ ns}}}$$

Hold calculations

FF 1 : Max clk path delay = 2 ns + FF hold = 2 ns + 3 ns = 5 ns

Min data path delay = 5 ns

$$\therefore \text{Hold time at B} = 5 \text{ ns} - 5 \text{ ns} = \underline{\underline{0}}$$

FF 2 :

For B : Max clk path delay = 2 ns + 2 ns = 4 ns + FF hold = 4 ns + 3 ns = 7 ns

Min data path delay = 5 ns + 2 ns + 4 ns + 4 ns = 15 ns

$$\therefore \text{Hold time at B} = 7 \text{ ns} - 15 \text{ ns} = -8 \text{ ns}$$

For A : Min data path delay = 4 ns

$$\therefore \text{Hold time at A} = 7 \text{ ns} - 4 \text{ ns} = \underline{\underline{3 \text{ ns}}}$$

For E input: Min data path delay = $2\text{ ns} + 4\text{ ns} + 4\text{ ns}$
 $= 10\text{ ns}$
 $\therefore$ Hold time at E = $7\text{ ns} - 10\text{ ns} = \underline{-3\text{ ns}}$

Operating freq calculation

Path	T Min
$U_1 - FF_1$	$8\text{ ns} + 4\text{ ns} = 12\text{ ns}$
$U_1 - U_2 - U_3 - U_4 - FF_2$	$8\text{ ns} + 4\text{ ns} + 7\text{ ns} + 6\text{ ns} + 4\text{ ns} = 29\text{ ns}$
$U_4 - FF_2$	$6\text{ ns} + 4\text{ ns} = 10\text{ ns}$
$U_2 - U_3 - U_4 - FF_2$	$4\text{ ns} + 7\text{ ns} + 6\text{ ns} + 4\text{ ns} = 21\text{ ns}$
$U_5 - U_6$	$5\text{ ns} + 8\text{ ns} = 13\text{ ns}$
$FF_1 - U_3 - U_4 - FF_2$	$8\text{ ns} + 7\text{ ns} + 6\text{ ns} + 4\text{ ns} = 25\text{ ns}$
$U_8 - U_9 - FF_2 - U_6$	$2\text{ ns} + 2\text{ ns} + 8\text{ ns} + 8\text{ ns} = 20\text{ ns}$

$$\therefore \text{Maximum frequency} = \frac{1}{29\text{ ns}} = \underline{34.48\text{ MHz}}$$

Question 5:

For path from A to X

$$H = \frac{60}{14}$$

$$B = 3 \times 2 = 6$$

$$G = \frac{7}{3} \times \frac{5}{3} \times \frac{5}{3} = \frac{175}{27}$$

$$\therefore F = GBH = \frac{60}{14} \times 6 \times \frac{175}{27} = 166.67$$

$$f = (166.67)^{1/3} = 5.50$$

$$\therefore D = Nf + P$$

$$= 3 \times 5.50 + 3 + 2 + 3 = 24.5$$

$$\therefore D = 24.5 \times 3RC = 24.5 \times 6\text{ ps} = \underline{147\text{ ps}}$$

$$f = g \times \frac{C_{out}}{C_{in}}$$

$$C_{in} = \frac{g \times C_{out}}{f} = \frac{\frac{5}{3} \times 60}{5.50} = 18.18$$

$$\therefore 5\lambda = 18.18$$

$$\lambda = 3.64$$

$$\text{NMOS: } 10.91$$

$$\text{PMOS: } 7.27$$

F04:

$$H = 4$$

$$d = gh + P$$

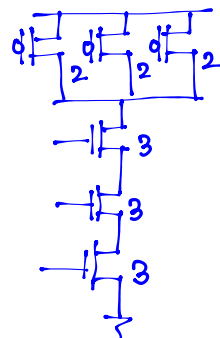
$$= 1 \times 4 + 1 = 5$$

$$5\tau = 30\text{ ps}$$

$$\tau = 6\text{ ps}$$

$$3RC = 6\text{ ps}$$

$$RC = 2\text{ ps}$$



$$C_{in} = \frac{g \times C_{out}}{f} = \frac{\frac{5}{3} \times (18 \cdot 18 + 18 \cdot 18)}{5 \cdot 50} = 11 \cdot 02$$

$$5\lambda = 11 \cdot 02$$

$$\lambda = 2 \cdot 204$$

$$NMOS = 2 \cdot 20$$

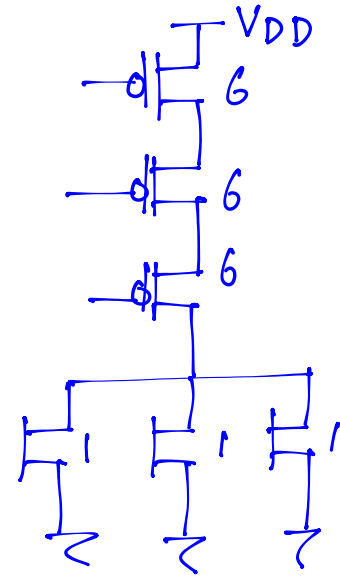
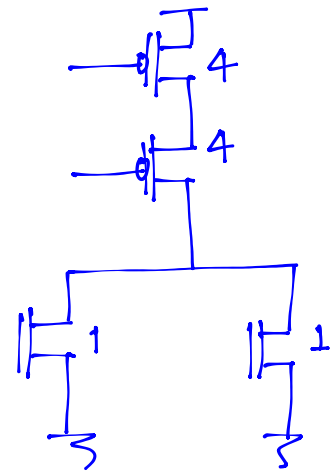
$$PMOS = 8 \cdot 82$$

$$C_{in} = \frac{\frac{7}{3} \times (11 \cdot 02 + 11 \cdot 02 + 11 \cdot 02)}{5 \cdot 50}$$

$$= 14$$

$$NMDS = 2$$

$$PMOS = 12$$



Path from A to Y

$$G = \frac{7}{3} \times \frac{5}{3} \times \frac{5}{3} \times \frac{5}{3} = \frac{875}{81}$$

$$B = 3 \times 2 = 6$$

$$H = \frac{75}{14}$$

$$F = GBH = \frac{875}{81} \times 6 \times \frac{75}{14} = 347 \cdot 22$$

$$f = (347 \cdot 22)^{1/4} = 4 \cdot 32$$

$$D = N_f + P = 4 \times 4 \cdot 32 + 3 + 2 + 3 + 2 = 27 \cdot 28$$

$$\therefore D = 27 \cdot 28 \times 3RC = 27 \cdot 28 \times 6 \text{ ps} = \underline{\underline{163 \cdot 68 \text{ ps}}}$$

$$C_{in} = \frac{g \times C_{out}}{f} = \frac{\frac{5}{3} \times 75}{4 \cdot 32} = 28 \cdot 94$$

$$5\lambda = 28 \cdot 94$$

$$NMOS = 5 \cdot 79$$

$$PMOS = 23 \cdot 15$$

$$C_{in} = \frac{\frac{5}{3} \times 28 \cdot 94}{4 \cdot 32} = 11 \cdot 17$$

$$5\lambda = 11.17$$

$$\lambda = 2.234$$

$$NMOS = 6.70$$

$$PMOS = 4.47$$

$$C_{in} = \frac{\frac{5}{3} \times (11.17 + 11.17)}{4.32} = 8.62$$

$$5\lambda = 8.62$$

$$NMOS = 1.72$$

$$PMOS = 6.90$$

$$C_{in} = \frac{\frac{7}{3} \times (8.62 + 8.62 + 8.62)}{4.32}$$

$$= 14$$

$$NMOS = 2$$

$$PMOS = 12$$

Question 6 :

$$\frac{I_{D6}}{I_{D5}} = \frac{\left(\frac{W}{L}\right)_6}{\left(\frac{W}{L}\right)_5} = \frac{\left(\frac{W}{L}\right)_6}{2\left(\frac{W}{L}\right)_6} = \frac{1}{2}$$

$$2I_{D6} = I_{D5}$$

$$PD = \frac{1}{2} (I_{D6} + I_{D5}) \times V_{DD}$$

$$0.54 \times 10 = 3I_{D6} \times 1.8$$

$$I_{D6} = 100 \mu A$$

$$I_{D5} = 200 \mu A$$

$$Gain = g_{m1} (r_{o2} \parallel r_{o4})$$

$$= \frac{2I_{D1}'}{V_{OV1}} \left(\frac{1}{(\lambda_n + \lambda_p) I_{D1}'} \right)$$

$$80 = \frac{2}{V_{OV1}} \frac{1}{(0.19 + 0.057)}$$

$$V_{OV1} = 0.1 V$$

$$I_{D5} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{OV})^2$$

$$200 \times 10^{-6} = \frac{1}{2} \times 300 \times 10^{-6} \times \frac{W}{L} \times (0.1)^2$$

$$\left(\frac{W}{L}\right)_5 = 133.33$$

$$I_{D1} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{OV})^2$$

$$100 \times 10^{-6} = \frac{1}{2} \times 300 \times 10^{-6} \times \frac{W}{L} \times (0.1)^2$$

$$\left(\frac{W}{L}\right)_{1,2} = 66.67$$

$$I_{D3} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{OV})^2$$

$$100 \times 10^{-6} = \frac{1}{2} \times 60 \times 10^{-6} \times \frac{W}{L} \times (0.1)^2$$

$$\left(\frac{W}{L}\right)_{3,4} = 333.33$$

$$V_{in,CM,min} = V_{SS} + V_{OV5} + V_{GS1}$$

$$= -0.9 + 0.1 + 0.48$$

$$= -0.32 V$$

$$V_{in,CM,max} = V_{DD} - |V_{GS3}| + V_{th1}$$

$$= 0.9 - 0.6 + 0.38$$

$$= 0.68 V$$

$$V_{out\ max} = V_{DD} - |V_{ov4}| = 0.9 - 0.1 = 0.8\text{ V}$$

$$V_{out\ min} = V_{SS} + V_{ov5} + V_{ov2} = -0.9 + 0.1 + 0.1 = -0.7\text{ V}$$

$$\therefore \text{Output voltage swing} = V_{out\ max} - V_{out\ min} \\ = 0.8 + 0.7 = \underline{1.5\text{ V}}$$

$$\beta = \frac{R_2}{R_1 + R_2}$$

$$A_{f\ ideal} = \frac{1}{\beta} = \frac{R_1 + R_2}{R_2} = 20$$

$$\frac{20\text{ M}\Omega}{R_2} = 20$$

$$R_2 = 1\text{ M}\Omega$$

$$R_1 = 19\text{ M}\Omega$$

$$g_{m1} = \frac{2I_D}{V_{ov1}} = \frac{2 \times 100 \times 10^{-6}}{0.1} = 2\text{ mS}$$

$$r_{o2} = \frac{1}{\lambda_n I_D} = \frac{1}{0.19 \times 100 \times 10^{-6}} = 52.63\text{ K}\Omega$$

$$r_{o4} = \frac{1}{\lambda_p I_D} = \frac{1}{0.057 \times 100 \times 10^{-6}} = 175.44\text{ K}\Omega$$

$$\begin{aligned} \text{Gain} &= 2 \times 10^{-3} \times (52.63\text{ K} \parallel 175.44\text{ K} \parallel 20\text{ M}) \\ &= 2 \times 10^{-3} \times (40.48\text{ K} \parallel 20\text{ M}) \\ &= 2 \times 10^{-3} \times 40.40 \times 10^3 = 80\text{ V/V} \end{aligned}$$

..... No loading effect on gain

$$\beta A = \frac{1}{20} \times 80 = 4$$

$$\therefore A_f = \frac{A}{1 + \beta A} = \frac{80}{1 + 4} = 16\text{ V/V}$$

$$\text{Output impedance} = \frac{52.63\text{ K} \parallel 175.44\text{ K} \parallel 20\text{ M}}{1 + \beta A}$$

$$= \frac{40.40\text{ K}}{1 + 4} = \underline{8.08\text{ K}\Omega}$$